

Residual Complements for Compression-Induced Macro-Closure Failure

Zihe Zhao

Independent Researcher
ybing2093@gmail.com
ORCID: 0009-0008-4692-4408

May 2026

Abstract

Coarse-graining and compression are often described as operations that discard information. This paper isolates a sharper closure mechanism: compression may hide distinctions inside fibers, and some of these hidden distinctions may remain dynamically active by later re-entering the declared macro-readout. Given a microscopic state space $\mathcal{X}$, a compression map $\pi : \mathcal{X} \rightarrow \mathcal{M}$, and a microscopic evolution map $T : \mathcal{X} \rightarrow \mathcal{X}$, a closed macro-dynamics $F : \mathcal{M} \rightarrow \mathcal{M}$ exists exactly when every compression fiber is future-constant under the one-step macro-readout $\pi \circ T$. This closure condition is standard in quotient/factor dynamics and has a stochastic analogue in Markov lumpability; here it is used as the entry point for a residual-complement diagnostic.

When future-constancy fails, the fiber-hidden distinction is identified as a feedback-active residual. The paper classifies fiber residuals by their induced variation under a declared future-readout map, separates candidate residual complement coordinates from closure-restoring residual complement coordinates, and states the congruence condition required for an enlarged macro-state $z = (\pi(x), c(x))$ to restore exact closure. The construction can also be iterated: when residual variation inside enlarged fibers crosses a declared closure threshold, further complement coordinates may be required, producing a recursive residual-complement hierarchy.

Finally, the paper defines interface-locked compensated macro-stability: a regime in which internal residual complement channels remain active while their projected macro-readout stays stable up to a declared threshold ε . This projected compensation is not an exact closure theorem unless the enlarged state is equipped with a closed dynamics satisfying the required congruence condition. The result is a closure-diagnostic module for determining when a chosen macro-readout is structurally insufficient, when hidden distinctions must be promoted into complement coordinates, and how stable projected behavior may coexist with active internal residual dynamics. Phase-transition-like examples are used as schematic instances of this complement-closure pattern, not as replacements for thermodynamic or renormalization-group theories.

Keywords: compression; coarse-graining; residual complements; macro-closure; projected stability; quotient dynamics; Markov lumpability; future-constancy; threshold crossing; declared interface; structural gaps

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1 Introduction

Compression is indispensable in the analysis of multi-level systems. A microscopic configuration is replaced by a macro-state, a detailed trajectory by an endpoint, or a high-dimensional distribution by a small set of reportable observables. Such compression is often useful, but its usefulness does not by itself imply macro-level closure.

The structural issue addressed in this note is the following. A compression map may hide distinctions inside fibers. These hidden distinctions may be irrelevant for some tasks, but they may also remain dynamically active: two microscopic states with the same macro-report may evolve into states with different future macro-reports. In that case, the macro-state is not exactly closed under the declared evolution.

This paper proposes a minimal formal structure for this situation. The main claims are:

- (i) Exact macro-closure exists if and only if compression fibers are future-constant under the one-step macro-readout.
- (ii) Failure of future-constancy identifies feedback-active residuals.
- (iii) Fiber residuals have different functional types: some are inert under the declared future readout, some are closure-breaking, some are separated by a candidate complement coordinate, some become closure-relevant under a closure-restoring extension, and some participate in compensated projected stability.
- (iv) Feedback-active residuals may be promoted to candidate residual complement coordinates; only those satisfying the required congruence condition restore enlarged macro-closure.
- (v) When residual variation crosses a declared closure threshold after extension, the construction can be iterated, yielding a recursive hierarchy of residual-complement macro-systems.
- (vi) Multiple residual complement channels may compensate under a declared interface, producing a stable or quasi-stable projected macro-readout while internal residual activity persists; this projected stability is not, by itself, a closure theorem.

The first item is a closure criterion familiar from quotient or factor dynamics, with a stochastic analogue in Markov lumpability. The novelty of the present note does not lie in treating this

condition as an isolated result. Rather, the closure criterion is used as a diagnostic trigger: once fiber future-constancy fails, the hidden within-fiber distinction is classified according to its future-readout role, promoted only when it is feedback-active under the declared package, and treated as closure-restoring only when the enlarged compression satisfies the required congruence condition. The contribution is therefore the residual-complement mechanism built around closure failure, not the bare closure condition alone.

The note is intended as a standalone formal module. It is compatible with viewpoint/package-locked structural-gap diagnostics, endpoint-fiber diagnostics, and accumulated residual book-keeping, but it does not identify these objects by default. Its role is narrower: it describes how compression-hidden residuals can become structural complements in an enlarged macro-system.

A guiding distinction used throughout the paper is the distinction between exact closure and report-level stability. Exact closure is a single-valued dynamics condition on the compressed state. Report-level closure or stability is a thresholded statement under a declared package and need not imply exact closure. Likewise, compensated projected stability is treated as a report-level phenomenon unless the enlarged state is also shown to satisfy an exact closure condition.

2 Relation to Existing Closure and Framework-Level Objects

This section positions the present construction without re-proving earlier work.

Declared interfaces and package locking. All report-level claims in this note are made relative to a declared interface or package $\mathcal{V}$. The package specifies what is visible, which comparisons are licensed, what counts as an admissible reconstruction, and which threshold ε determines report-level negligibility.

Relation to quotient dynamics and lumpability. The exact closure condition used in Proposition 3.2 is the standard well-definedness condition for a quotient or factor dynamics: the evolution must respect the equivalence relation induced by the compression map. In the stochastic finite-state setting, the corresponding condition is Markov lumpability: all microscopic representatives inside the same macro-fiber must induce the same macro-transition law [5, 6]. The present paper does not claim these closure criteria as new. It uses them as a diagnostic boundary: when they fail, the question becomes which fiber-hidden distinctions are functionally active, which of them should be promoted as residual complement coordinates, and which promoted coordinates actually restore closure under the declared congruence condition.

Relation to projection and memory formalisms. Projection-based approaches often represent the effect of unresolved variables through memory, noise, or orthogonal-dynamics terms [7, 8]. The present note takes a complementary diagnostic route: it asks when an unresolved distinction should instead be promoted into an explicit complement coordinate so that an enlarged macro-state may satisfy a closure congruence condition.

Fiber residuals. The present construction uses the same basic geometric idea as endpoint/fiber diagnostics: if a compression map identifies several fine-grained states, the resulting fiber may still contain distinctions relevant to future reports. The present paper adds a dynamical question: which hidden distinctions become functionally active under the declared future-readout map?

Fixed-slice gaps versus accumulated quantities. A fixed-slice residual, such as a compression-reconstruction gap, should not be identified with an accumulated residual along an evolution parameter. In this paper, compensation occurs at the level of residual complement

channels under a declared projection, not as cancellation of a nonnegative accumulated bookkeeping quantity.

Extension principle. When a macro-report fails to be single-valued with respect to the declared future query, the appropriate response is not to force closure, but to enlarge the reporting structure. In the terminology used here, this enlargement is implemented by residual complement coordinates.

No new Delta-type primitive. This note does not introduce a new Δ -type scalar as a primitive object. A gap or residual magnitude may be defined after a discrepancy and a report map are declared, but the primary object here is the closure mechanism by which compression-hidden distinctions become feedback-active residual complements.

Declared complement package. For the rest of this note, the minimal package underlying complement-closure statements is

$$\mathcal{V}_{\text{comp}} := (\mathcal{X}_{\text{adm}}, \pi, T, R_{\mathcal{V}}, D_{\mathcal{V}}, \varepsilon, k, \mathcal{C}_{\text{ext}}), \quad (1)$$

where $\mathcal{X}_{\text{adm}} \subseteq \mathcal{X}$ is the admissible or reachable domain, $\pi : \mathcal{X} \rightarrow \mathcal{M}$ is the declared compression map, T is the declared evolution on the required horizon, $R_{\mathcal{V}}$ is the report map, $D_{\mathcal{V}}$ is the licensed report discrepancy, $\varepsilon > 0$ is the reporting threshold, k is the declared future horizon, and $\mathcal{C}_{\text{ext}}$ is the admissible class of complement extensions. Unless otherwise stated, closure, residual activity, threshold crossing, and compensation claims are evaluated on $\mathcal{X}_{\text{adm}}$ under this package. We assume $T^j(\mathcal{X}_{\text{adm}}) \subseteq \mathcal{X}_{\text{adm}}$ for the horizons j used in the corresponding claim; if this condition is not declared, the corresponding finite-horizon report is not licensed under $\mathcal{V}_{\text{comp}}$.

Relation to the earlier $H \rightarrow H^+$ language. Relative to the earlier structural-extension language, the present paper supplies an explicit coordinate-level implementation of the carrier extension: when closure restoration is licensed, the complement coordinate c carries within-fiber information required for closure under the declared future query. This does not identify the present complement coordinate with endpoint–path residuals, fixed-slice fiber diameters, or accumulated residual bookkeeping.

3 Compression Fibers and Macro-Closure

Let $\mathcal{X}$ be a microscopic state space and let

$$\pi : \mathcal{X} \rightarrow \mathcal{M} \quad (2)$$

be a compression or coarse-graining map into a macro-state space $\mathcal{M}$. For $m \in \mathcal{M}$, define the compression fiber

$$\text{Fib}_{\pi}(m) := \pi^{-1}(m) = \{x \in \mathcal{X} : \pi(x) = m\}. \quad (3)$$

Let

$$T : \mathcal{X} \rightarrow \mathcal{X} \quad (4)$$

be the microscopic evolution map.

Definition 3.1 (Exact macro-closure). *The compressed description $(\mathcal{M}, \pi)$ is exactly closed under T on the admissible domain $\mathcal{X}_{\text{adm}}$ if there exists a map*

$$F : \mathcal{M} \rightarrow \mathcal{M} \quad (5)$$

such that

$$\pi \circ T = F \circ \pi \quad \text{on } \mathcal{X}_{\text{adm}}. \quad (6)$$

In this case, F is called the induced exact macro-dynamics on the declared domain.

Proposition 3.2 (Exact macro-closure iff fiber future-constancy). *An exact closed macro-dynamics $F : \mathcal{M} \rightarrow \mathcal{M}$ satisfying (6) on $\mathcal{X}_{\text{adm}}$ exists if and only if, for all $x, x' \in \mathcal{X}_{\text{adm}}$,*

$$\pi(x) = \pi(x') \implies \pi(Tx) = \pi(Tx'). \quad (7)$$

Proof. Assume F exists. If $\pi(x) = \pi(x')$, then

$$\pi(Tx) = F(\pi(x)) = F(\pi(x')) = \pi(Tx'),$$

so (7) holds on $\mathcal{X}_{\text{adm}}$.

Conversely, assume (7) on $\mathcal{X}_{\text{adm}}$. Define F on $\pi(\mathcal{X}_{\text{adm}}) \subseteq \mathcal{M}$ by

$$F(\pi(x)) := \pi(Tx). \quad (8)$$

The condition (7) guarantees that this definition is independent of the representative x chosen from the fiber. Extend F arbitrarily outside $\pi(\mathcal{X}_{\text{adm}})$ if necessary. Then $\pi \circ T = F \circ \pi$ on $\mathcal{X}_{\text{adm}}$. $\square$

Remark 3.3 (Meaning of failure). *Failure of (7) does not mean that the macro-variable is useless. It means that the macro-variable is insufficient for closed prediction under the declared evolution and readout. The missing information is localized inside compression fibers.*

4 Future-Readout Maps and Feedback-Active Residuals

The one-step condition in Proposition 3.2 is the minimal closure test. For applications, a residual may be invisible after one step but visible over a longer declared horizon. We therefore introduce a future-readout map.

Let $R_V : \mathcal{M} \rightarrow \mathcal{R}$ be a declared macro-report map and let D_V be a discrepancy on $\mathcal{R}$. For a horizon $k \geq 1$ licensed by $\mathcal{V}_{\text{comp}}$, define, for $x \in \mathcal{X}_{\text{adm}}$,

$$B_{V,k}(x) := (R_V(\pi(Tx)), R_V(\pi(T^2x)), \dots, R_V(\pi(T^kx))). \quad (9)$$

When $R_V = \text{id}_{\mathcal{M}}$, this records the declared macro-trajectory up to horizon k .

Definition 4.1 ($\mathcal{V}$ -report future-readout closure). *Fix the declared package $\mathcal{V}_{\text{comp}}$ and a product discrepancy $D_V^{(k)}$ on future-readout strings. The compression π is $\mathcal{V}$ -report closed at horizon k and threshold ε if, for all $x, x' \in \mathcal{X}_{\text{adm}}$,*

$$\pi(x) = \pi(x') \implies D_V^{(k)}(B_{V,k}(x), B_{V,k}(x')) < \varepsilon. \quad (10)$$

This is a thresholded report-level condition. Exact macro-closure from Proposition 3.2 is the stronger one-step case in which the macro-readout is required to be single-valued, rather than merely indistinguishable below ε .

Definition 4.2 (Fiber residual). *Let $x, x' \in \mathcal{X}_{\text{adm}}$ satisfy $\pi(x) = \pi(x')$. The pair (x, x') is called a fiber residual pair under π if $x \neq x'$. It is a distinction hidden by the compression map.*

Definition 4.3 (One-step feedback-active residual). *A fiber residual pair (x, x') is one-step feedback-active under (π, T) if*

$$\pi(x) = \pi(x') \quad \text{but} \quad \pi(Tx) \neq \pi(Tx'). \quad (11)$$

Equivalently, the residual is invisible at the current macro-readout but visible in the next macro-readout.

Definition 4.4 (Horizon- k feedback-active residual). *A fiber residual pair (x, x') is feedback-active under (π, T, R_V, k) if*

$$\pi(x) = \pi(x') \quad \text{but} \quad B_{V,k}(x) \neq B_{V,k}(x'). \quad (12)$$

It is ε -active if the discrepancy between the two future-readout strings is at least ε under the declared product discrepancy.

Corollary 4.5. *A compressed macro-description is not exactly closed under T if and only if there exists at least one one-step feedback-active residual pair.*

Proof. Immediate from Proposition 3.2. □

Remark 4.6 (No information-conservation claim). *The term “residual” should not be read as a universal information-conservation statement. It means that a distinction hidden by the declared compression remains dynamically relevant within the declared system boundary and future readout.*

5 Functional Types of Fiber Residuals

The existence of a fiber difference is not yet a structural conclusion. Its role is determined by how it behaves under the declared compression, evolution, report map, and possible macro-extension.

Definition 5.1 (Inert residual). *A fiber residual pair (x, x') is inert under (π, T, R_V, k) if*

$$\pi(x) = \pi(x') \quad \text{and} \quad B_{V,k}(x) = B_{V,k}(x'). \quad (13)$$

Such a distinction is hidden by the current compression and does not affect the declared future-readout over the horizon k .

Definition 5.2 (Closure-breaking residual). *A fiber residual pair (x, x') is closure-breaking for (π, T) if*

$$\pi(x) = \pi(x') \quad \text{and} \quad \pi(Tx) \neq \pi(Tx'). \quad (14)$$

Equivalently, it violates the fiber future-constancy condition (7).

Definition 5.3 (c -separated residual pair). *Let $c : \mathcal{X} \rightarrow \mathcal{E}$. A fiber residual pair (x, x') is c -separated if*

$$\pi(x) = \pi(x') \quad \text{but} \quad c(x) \neq c(x'). \quad (15)$$

This is a local separation property of the coordinate c ; by itself it does not imply that c restores macro-closure.

Definition 5.4 (Closure-relevant residual pair). *Let $c : \mathcal{X} \rightarrow \mathcal{E}$ and $\pi^+ = (\pi, c)$. A c -separated residual pair (x, x') is closure-relevant for the declared extension π^+ if c is part of a closure-restoring residual complement coordinate in the sense of Definition 6.2. Thus closure-relevance is evaluated relative to the enlarged compression and its declared congruence condition, not from pairwise separation alone.*

Definition 5.5 (Compensatory residual). *A residual complement state $\eta_t = c(x_t)$ is compensatory on a time window I if it changes on I while the declared macro-readout of the enlarged state remains ε -stable. The precise report-level condition is stated in Definition 8.1.*

Remark 5.6 (Functional status is package-relative). *The functional status of a fiber residual is relative to (π, T, R_V, k) and, when an extension is used, to $\pi^+ = (\pi, c)$. The same microscopic distinction may be inert under one declared readout and closure-breaking under another.*

6 Residual Complement Coordinates

When feedback-active residuals exist, one can attempt to enlarge the macro-state by adding residual complement coordinates.

Definition 6.1 (Candidate residual complement coordinate). *A map*

$$c : \mathcal{X} \rightarrow \mathcal{E} \quad (16)$$

is called a candidate residual complement coordinate for (π, T) on $\mathcal{X}_{\text{adm}}$ if the enlarged compression map

$$\pi^+ : \mathcal{X} \rightarrow \mathcal{M}^+, \quad \pi^+(x) := (\pi(x), c(x)) \quad (17)$$

is admitted by the declared extension class $\mathcal{C}_{\text{ext}}$ and either separates at least one declared feedback-active fiber residual or belongs to a licensed family declared to carry compression-hidden information relevant to the declared future-readout. This definition makes no closure claim.

Definition 6.2 (Closure-restoring residual complement coordinate). *A candidate residual complement coordinate $c : \mathcal{X} \rightarrow \mathcal{E}$ is closure-restoring for (π, T) on $\mathcal{X}_{\text{adm}}$ if the enlarged compression map $\pi^+ = (\pi, c)$ admits an exact closed enlarged macro-dynamics $F^+ : \mathcal{M}^+ \rightarrow \mathcal{M}^+$ satisfying*

$$\pi^+ \circ T = F^+ \circ \pi^+ \quad \text{on } \mathcal{X}_{\text{adm}}. \quad (18)$$

Proposition 6.3 (Closure restoration iff enlarged-fiber congruence). *Let $c : \mathcal{X} \rightarrow \mathcal{E}$ be a candidate residual complement coordinate for (π, T) on $\mathcal{X}_{\text{adm}}$, and set $\pi^+ = (\pi, c)$. The following are equivalent:*

(i) *c is closure-restoring for (π, T) on $\mathcal{X}_{\text{adm}}$.*

(ii) *For all $x, x' \in \mathcal{X}_{\text{adm}}$,*

$$\pi(x) = \pi(x') \text{ and } c(x) = c(x') \implies \pi(Tx) = \pi(Tx') \text{ and } c(Tx) = c(Tx'). \quad (19)$$

In this case, the induced enlarged macro-dynamics is defined on $\pi^+(\mathcal{X}_{\text{adm}})$ by

$$F^+(\pi^+(x)) := \pi^+(Tx), \quad (20)$$

and may be extended arbitrarily outside $\pi^+(\mathcal{X}_{\text{adm}})$ if needed.

Proof. Apply Proposition 3.2 to the enlarged compression map π^+ . Condition (19) is exactly the fiber future-constancy condition for π^+ , and therefore is equivalent to the existence of an exact closed enlarged macro-dynamics satisfying Definition 6.2. $\square$

Remark 6.4 (Trivial full-state closure and nontrivial complements). *In a deterministic setting, exact closure can always be restored trivially by taking a complement coordinate that preserves the full microscopic state, for example $c(x) = x$. Such an extension is not the diagnostic target of this paper. The nontrivial content of residual-complement closure appears only after an admissible extension class $\mathcal{C}_{\text{ext}}$, a reporting interface, and, when needed, a cost, minimality, or compression discipline have been declared. A residual complement coordinate is therefore not merely any coordinate that restores closure, but a licensed carrier of the feedback-active distinctions required by the declared macro-readout.*

Lemma 6.5 (Closure-restoring complements separate closure-breaking pairs). *Let $c : \mathcal{X} \rightarrow \mathcal{E}$ be closure-restoring for (π, T) on $\mathcal{X}_{\text{adm}}$. If $x, x' \in \mathcal{X}_{\text{adm}}$ satisfy*

$$\pi(x) = \pi(x') \quad \text{and} \quad \pi(Tx) \neq \pi(Tx'), \quad (21)$$

then

$$c(x) \neq c(x'). \quad (22)$$

Proof. If $c(x) = c(x')$, then $\pi^+(x) = \pi^+(x')$. Since c is closure-restoring, the enlarged compression satisfies the congruence condition in Proposition 6.3. Hence $\pi(Tx) = \pi(Tx')$, contradicting the assumed closure-breaking pair. $\square$

Remark 6.6 (Interpretation). *A closure-restoring residual complement coordinate c is not an arbitrary extra variable. It extracts enough feedback-active fiber information to make the enlarged macro-description exactly closed under the declared dynamics. A candidate coordinate need not restore closure unless the congruence condition in Proposition 6.3 holds.*

Coordinate map versus complement state. Throughout the rest of the paper, c denotes the residual complement coordinate map, while

$$\eta_t := c(x_t) \quad (23)$$

denotes the residual complement state realized along a trajectory $x_t = T^t x_0$. When the complement map has several components, we write

$$c = (c_1, \dots, c_r), \quad \eta_t = (\eta_{1,t}, \dots, \eta_{r,t}), \quad \eta_{i,t} := c_i(x_t). \quad (24)$$

Thus η_t is the activated residual-complement channel state induced by c along the trajectory, not a new primitive object independent of c .

Definition 6.7 (Minimal future-readout quotient). *Fix (π, T, R_V, k) . Define an equivalence relation on $\mathcal{X}_{\text{adm}}$ by*

$$x \sim_{V,k} x' \iff \pi(x) = \pi(x') \text{ and } B_{V,k}(x) = B_{V,k}(x'). \quad (25)$$

The horizon- k quotient coordinate

$$c_{V,k}^*(x) := [x]_{\sim_{V,k}} \quad (26)$$

separates exactly those fiber differences that change the declared future-readout over horizon k .

Remark 6.8 (Finite-horizon separation is not closure). *The coordinate $c_{V,k}^*$ is a minimal horizon- k separator. It is not, by itself, a closure-restoring complement coordinate unless the induced equivalence relation is T -invariant on $\mathcal{X}_{\text{adm}}$, or unless the construction is replaced by an infinite-horizon or stable-refinement version. Closure still requires the congruence condition in Proposition 6.3.*

Remark 6.9 (No demand to preserve all micro-information). *The construction does not require preservation of all microscopic distinctions. Only functionally active fiber differences, as measured by the declared future-readout, are candidates for promotion into complement coordinates; closure restoration still requires the relevant congruence condition.*

Coordinate representation: residual counter-terms

When the macro-state space has a vector-space or charted structure, the influence of residual complements can be written as a counter-term. This is an optional coordinate representation, not the coordinate-free core of the construction.

Assumption 6.10 (Local chart). *Assume that $\mathcal{M}$ is locally represented in a coordinate chart so that differences of macro-readouts are meaningful in a normed vector space.*

Let $m_t = \pi(x_t)$ and $\eta_t = c(x_t)$. Suppose the enlarged dynamics can be written locally as

$$m_{t+1} = \mathcal{F}(m_t, \eta_t). \quad (27)$$

Choose a baseline macro-map $F_0 : \mathcal{M} \rightarrow \mathcal{M}$. Define the residual complement effect by

$$C(m_t, \eta_t) := \mathcal{F}(m_t, \eta_t) - F_0(m_t), \quad (28)$$

whenever this expression is meaningful in the declared chart.

Definition 6.11 (Residual counter-term). *The term $C(m, \eta)$ is a residual counter-term if it measures the reportable deviation in the macro-update induced by the feedback-active residual coordinate η relative to a baseline macro-update $F_0(m)$.*

Remark 6.12 (Coordinate dependence). *The counter-term representation is chart-dependent. The coordinate-free object is the failure of closure under π and the restoration of closure under π^+ .*

7 Threshold-Crossing and Recursive Complement Closure

The preceding sections describe a single closure attempt. The same mechanism can occur recursively. After a complement coordinate has been added, the enlarged macro-system may still have its own compression fibers and its own feedback-active residuals.

Let

$$\pi_n : \mathcal{X} \rightarrow \mathcal{M}_n \quad (29)$$

be the macro-interface at level n , equipped with report map $R_n : \mathcal{M}_n \rightarrow \mathcal{R}_n$, licensed product discrepancy $D_n^{(k_n)}$ on $\mathcal{R}_n^{k_n}$, threshold $\varepsilon_n > 0$, and horizon k_n . Define the level- n future-readout map

$$B_{n,k_n}(x) := (R_n(\pi_n(Tx)), \dots, R_n(\pi_n(T^{k_n}x))). \quad (30)$$

For $m \in \pi_n(\mathcal{X}_{\text{adm}})$, define the level- n fiber variation

$$\rho_{n,k_n}(m) := \sup_{x, x' \in \text{Fib}_{\pi_n}(m) \cap \mathcal{X}_{\text{adm}}} D_n^{(k_n)}(B_{n,k_n}(x), B_{n,k_n}(x')), \quad (31)$$

whenever the displayed discrepancy is licensed under the declared package.

Remark 7.1 (Relation to fiber-diameter diagnostics). *The quantity $\rho_{n,k_n}(m)$ is the dynamic future-readout analogue of a fiber diameter. In fixed-slice endpoint diagnostics, a fiber diameter measures the maximal discrepancy among outputs associated with states or histories that share the same endpoint. Here the endpoint is the level- n macro-report $\pi_n(x) = m$, and the readout being compared is the declared future-readout map B_{n,k_n} . Thus $\rho_{n,k_n}(m)$ plays the same structural role as a fiber diameter, but on a dynamical readout window. When $\rho_{n,k_n}(m) \geq \varepsilon_n$, the diagnostic diameter is not merely a measure of non-sufficiency; it becomes a trigger for structural extension, namely the promotion of a residual complement coordinate c_n . This is the point at which a fiber-diameter diagnostic becomes an extension signal under the declared package [3].*

Definition 7.2 (Threshold-crossing residual). *A level- n fiber over $m \in \mathcal{M}_n$ contains a threshold-crossing residual if*

$$\rho_{n,k_n}(m) \geq \varepsilon_n. \quad (32)$$

When this occurs, the residual is reportably active at level n under the declared interface.

Definition 7.3 (Recursive residual-complement extension). *A recursive residual-complement extension is a sequence*

$$\pi_0 \longrightarrow \pi_1 \longrightarrow \pi_2 \longrightarrow \dots \quad (33)$$

where

$$\pi_{n+1}(x) := (\pi_n(x), c_n(x)) \quad (34)$$

for a complement coordinate c_n introduced to separate threshold-crossing residuals at level n .

Remark 7.4 (No termination, uniqueness, or minimality claim). *The recursive construction does not imply finite termination, uniqueness of complement coordinates, or minimality of the resulting hierarchy unless additional admissibility and minimality clauses are declared. Each step introduces a candidate extension triggered by threshold-crossing residual variation; closure restoration at the next level still requires the corresponding congruence condition.*

Remark 7.5 (Fractal-like, not automatically fractal). *The recursive construction creates nested fiber refinements*

$$\text{Fib}_{\pi_0} \supseteq \text{Fib}_{\pi_1} \supseteq \text{Fib}_{\pi_2} \supseteq \dots \quad (35)$$

whenever the maps are formed by adding coordinates. This gives a fractal-like residual hierarchy only if a declared rescaling or recurrence law identifies similar residual-complement patterns across levels. Without such a rescaling law, the structure is recursive but not automatically a fractal.

8 Interface-Locked Compensated Projected Stability

In this note, compensated projected stability means projected report-level stability under the declared interface while residual complement channels remain active. This condition alone is not a closure theorem. Exact compensated macro-closure requires, in addition, an enlarged state equipped with closed dynamics satisfying the required congruence condition.

Let the enlarged state be

$$z_t = (m_t, \eta_t) \in \mathcal{M}^+. \quad (36)$$

Let

$$R_{\mathcal{V}}^+ : \mathcal{M}^+ \rightarrow \mathcal{R} \quad (37)$$

be a declared enlarged-state report map associated with a declared interface or package $\mathcal{V}$. The space $\mathcal{R}$ is equipped with a report-level discrepancy $D_{\mathcal{V}}$ and threshold $\varepsilon > 0$.

Definition 8.1 (Interface-locked compensated projected stability). *A trajectory $(z_t)_{t \geq 0}$ satisfies interface-locked compensated projected stability under $(R_{\mathcal{V}}^+, D_{\mathcal{V}}, \varepsilon)$ on a time window $I \subseteq \mathbb{N}$ if*

$$D_{\mathcal{V}}(R_{\mathcal{V}}^+(z_{t+1}), R_{\mathcal{V}}^+(z_t)) < \varepsilon \quad \text{for all } t, t+1 \in I, \quad (38)$$

while the residual complement coordinates remain active, i.e. there exists at least one $t \in I$ such that

$$\eta_{t+1} \neq \eta_t \quad (39)$$

or, in a metric version,

$$d_{\eta}(\eta_{t+1}, \eta_t) > 0. \quad (40)$$

Definition 8.2 (Channel-wise compensation). *Suppose $\eta = (\eta_1, \dots, \eta_N)$ and the residual complement effects decompose locally as*

$$C_{\text{tot}}(m, \eta) = \sum_{i=1}^N C_i(m, \eta_i). \quad (41)$$

Let $\|\cdot\|_{\text{int}}$ be a declared internal activity norm and let $\delta_{\text{int}} > 0$ be a declared internal activity threshold. The residual channels are compensated under the declared projection $\Pi_{\mathcal{V}}$ if

$$\sum_{i=1}^N \|C_i(m, \eta_i)\|_{\text{int}} \geq \delta_{\text{int}}, \quad (42)$$

while the projected report-level effect is negligible:

$$\left\| \Pi_{\mathcal{V}} \left(\sum_{i=1}^N C_i(m, \eta_i) \right) \right\|_{\mathcal{V}} < \varepsilon, \quad (43)$$

where $\|\cdot\|_{\mathcal{V}}$ is the report-level norm licensed by the declared interface. Equivalently, when the interface is expressed by a report map and discrepancy, condition (43) may be replaced by the corresponding $D_{\mathcal{V}}$ -inequality.

Remark 8.3 (No cancellation of accumulated residual bookkeeping). *Channel-wise compensation is not a claim that a nonnegative accumulated residual vanishes. It is a claim about projected complement effects under a declared interface. Gross residual activity and net report-level residual may differ.*

Remark 8.4 (Coordinate-level status of channel decomposition). *The decomposition $C_{\text{tot}} = \sum_i C_i$ is a coordinate-level representation licensed by the declared chart or local model. The coordinate-free claim is only that projected report-level variation may remain below ε while residual complement channels remain internally active.*

9 Example I: Markov Coarse-Graining

This section gives the stochastic analogue of Proposition 3.2. The deterministic map T is replaced by a Markov kernel K , and closure means the existence of a macro-kernel $\bar{K}(m, \cdot)$ whose value is independent of the chosen microscopic representative inside each fiber.

Let $\mathcal{X}$ be a finite microscopic state space, let $\pi : \mathcal{X} \rightarrow \mathcal{M}$ be a coarse-graining map, and let $K(x, \cdot)$ be a Markov transition kernel on $\mathcal{X}$. For each $x \in \mathcal{X}$, the induced one-step macro-transition law is the pushforward

$$\bar{K}_x(A) := K(x, \pi^{-1}(A)), \quad A \subseteq \mathcal{M}. \quad (44)$$

A closed Markov macro-kernel $\bar{K}(m, \cdot)$ exists if and only if the pushforward law is constant on compression fibers:

$$\pi(x) = \pi(x') \implies \bar{K}_x = \bar{K}_{x'}. \quad (45)$$

Condition (45) is the stochastic version of fiber future-constancy. When it holds, one defines $\bar{K}(m, \cdot) := \bar{K}_x(\cdot)$ for any representative $x \in \pi^{-1}(m)$, and the definition is representative-independent. In finite-state Markov terminology, this is the corresponding lumpability condition under the declared coarse-graining map.

The residual-complement reading begins after this standard closure test fails. In that case, the non-constant pushforward laws do not merely show that the coarse Markov kernel is undefined; they localize which within-fiber distinctions are feedback-active relative to the declared macro-transition query. A complement coordinate c is then evaluated not by whether it separates some microscopic states, but by whether the enlarged fibers of (π, c) make the induced macro-transition law constant.

For a concrete example, take $\mathcal{X} = \{1, 2, 3, 4\}$ and $\mathcal{M} = \{A, B\}$ with

$$\pi(1) = \pi(2) = A, \quad \pi(3) = \pi(4) = B. \quad (46)$$

If

$$K(1, \{3\}) = 1, \quad K(2, \{4\}) = 1, \quad (47)$$

then both microscopic states in $\text{Fib}_\pi(A)$ move to macro-state B with probability one, so the pair $(1, 2)$ is not feedback-active relative to the one-step macro-readout. If instead

$$K(1, \{3\}) = 1, \quad K(2, \{1\}) = 1, \quad (48)$$

then

$$\pi(1) = \pi(2) = A \quad \text{but} \quad \bar{K}_1 = \delta_B \neq \delta_A = \bar{K}_2. \quad (49)$$

Hence the macro-state A is not closed: the hidden distinction inside $\text{Fib}_\pi(A)$ changes the future macro-transition law. The displayed pair can be separated by a residual complement coordinate, for example

$$c(1) = 0, \quad c(2) = 1. \quad (50)$$

These two values only separate the active pair $(1, 2)$. A complete stochastic closure statement requires values on all states and constancy of the induced transition law on every enlarged fiber. For instance, if the remaining transitions are declared by

$$K(3, \{3\}) = 1, \quad K(4, \{4\}) = 1, \quad (51)$$

and one sets

$$c(3) = 0, \quad c(4) = 1, \quad (52)$$

then each enlarged fiber of (π, c) is a singleton in this toy example, and the enlarged macro-kernel is closed.

10 Example II: Temperature Compression and Phase-Transition-Like Behavior

This example is illustrative rather than a thermodynamic derivation of phase transitions. It shows how a temperature-only readout can fail closure when phase composition or order-parameter-like structure becomes feedback-active. The example should therefore be read as a schematic complement-closure pattern, not as a full thermodynamic model.

This example should not be read as deriving the order parameter, free energy, critical exponents, or universality class of a thermodynamic transition. Its diagnostic target is different. Starting from a chosen macro-readout Θ , it asks whether Θ alone is closed under the declared heat-input protocol. When equal-temperature states differ in phase composition or order-parameter-like structure and therefore produce different future temperature responses, Θ is structurally insufficient as a closed macro-state. The complement coordinate ϕ is introduced not as a replacement for thermodynamic order-parameter theory, but as the residual carrier required to test whether the enlarged readout (Θ, ϕ) satisfies a closure condition.

Let $\mathcal{X}$ denote microscopic thermal states and let

$$\Theta : \mathcal{X} \rightarrow \mathbb{R}, \quad \Theta(x) = \text{temperature of } x \quad (53)$$

be the temperature readout. States with the same temperature may differ in phase fraction, latent heat storage, interface geometry, density fluctuations, or local molecular organization. Let

$$\mathcal{T}_Q : \mathcal{X} \rightarrow \mathcal{X} \quad (54)$$

denote the declared microscopic thermal evolution under a fixed heat-input or control protocol Q . The protocol Q is understood to include the controlled external conditions relevant to the readout, such as boundary conditions, pressure/volume constraints, heat-input schedule, or external fields when these are part of the declared experiment. A temperature-only description is closed under this protocol only if

$$\Theta(x) = \Theta(x') \implies \Theta(\mathcal{T}_Q x) = \Theta(\mathcal{T}_Q x'). \quad (55)$$

Near a transition window this condition can fail: two states with the same temperature may have different phase compositions and therefore respond differently to the same declared heat input.

Introduce a residual complement coordinate

$$\phi : \mathcal{X} \rightarrow [0, 1], \quad (56)$$

where ϕ denotes a phase fraction or an order-parameter-like coordinate. The enlarged macro-state is

$$z = (\Theta, \phi). \quad (57)$$

During a transition window, the report-level temperature may remain approximately stable,

$$|\Theta_{t+1} - \Theta_t| < \varepsilon, \quad (58)$$

while the complement coordinate changes,

$$|\phi_{t+1} - \phi_t| > 0. \quad (59)$$

Thus the temperature plateau is not interpreted as absence of internal change, but as an interface-locked compensated regime: under this declared interface, the temperature readout remains ε -stable while the complement coordinate changes. No thermodynamic closure theorem is asserted unless the enlarged state (Θ, ϕ) is equipped with a declared closed dynamics satisfying the relevant congruence condition.

11 Discussion

The construction separates six notions that are often conflated:

- (i) **Compression:** the map $\pi : \mathcal{X} \rightarrow \mathcal{M}$ hides distinctions inside fibers.
- (ii) **Residual activity:** some hidden distinctions remain relevant to future macro-readouts.
- (iii) **Functional residual type:** a fiber difference may be inert, closure-breaking, c -separated, closure-relevant, or compensatory depending on (π, T, R_V, k) and the chosen extension.
- (iv) **Candidate versus closure-restoring complement extension:** feedback-active residuals can be promoted to candidate macro-coordinates, but only congruence-satisfying extensions restore exact closure.
- (v) **Exact closure versus report-level closure:** exact closure requires single-valued macro-dynamics, whereas report-level closure is thresholded under a declared package.
- (vi) **Compensated stability:** the enlarged system may have active complement dynamics while the declared macro-readout remains stable up to threshold ε ; this projected stability is not itself a closure theorem.

The construction should therefore be read as a closure-extension module rather than as a new name for the closure condition itself. The closure condition tells us when a compressed macro-state is well-defined as a closed dynamics. The residual-complement construction begins when that condition fails: it identifies the active hidden distinctions, separates inert residuals from closure-breaking residuals, distinguishes merely separating coordinates from closure-restoring coordinates, and permits recursive extension when the enlarged description still leaves threshold-crossing fiber variation. This is the structural increment supplied by the present note.

Not every fiber-hidden distinction deserves promotion into a complement coordinate. Only distinctions that produce variation under the declared future-readout map are functionally active. Among them, only those needed for a declared congruence condition become closure-relevant. This prevents the construction from becoming a demand to preserve all microscopic information. Full-state embedding can restore closure trivially; the diagnostic target is a licensed, nontrivial complement inside the declared extension class.

The recursive extension in Section 7 also clarifies the relation between macro-level formation and threshold crossing. A macro-level is not only selected by compression; it may be dynamically generated when fiber-hidden distinctions cross the declared closure threshold and must be promoted into candidate complement coordinates. If the same residual-complement pattern recurs across levels under a declared rescaling law, the resulting hierarchy may be self-similar. Without such a law, the hierarchy remains recursive but not automatically fractal.

This suggests a structural reading of certain emergence and phase-transition-like phenomena. A macro-state may appear stable not because lower-level residuals vanish, but because residual complement channels are organized so that their projected report-level effect is stable or near-zero under a declared interface.

12 Conclusion

Compression does not by itself determine whether hidden distinctions are structurally irrelevant. The decisive condition is future-constancy on compression fibers. When fiber-hidden distinctions affect future macro-readouts, they are feedback-active residuals. Promoting such residuals to candidate complement coordinates yields an enlarged macro-system; closure is restored only when the enlarged compression satisfies the required congruence condition. When residual variation inside the enlarged fibers again crosses a declared threshold, the same construction can iterate, producing a recursive residual-complement hierarchy. When multiple residual complement channels compensate under a declared interface, a stable projected macro-readout can coexist with persistent internal residual activity, without implying exact closure by itself.

The minimal slogan is:

A macro-state is exactly closed only when all states in the same compression fiber have the same future macro-readout. When this fails, feedback-active residuals must either be reported as non-closure or promoted to candidate residual complement coordinates; closure restoration then requires a declared congruence condition on the enlarged fibers.

In this sense, the paper does not replace existing closure, lumpability, thermodynamic, or phase-transition theories. It supplies a package-relative residual-complement language for what happens when a chosen compression fails closure and must be extended.

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